

## RESEARCH ABSTRACT

Title: Efficacy and Safety of Metformin/Glipizide Combination Therapy in Adults with Type 2 Diabetes Inadequately Controlled on Metformin Monotherapy: A Randomized, Double-Blind, Placebo-Controlled Trial (Fictional Study, "GLIDE-7")

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### BACKGROUND

Patients with type 2 diabetes mellitus inadequately controlled on metformin monotherapy often require add-on therapy to achieve glycemic targets. This trial evaluated the fixed-dose combination metformin/glipizide against placebo added to existing metformin therapy.

### METHODS

GLIDE-7 was a 26-week, multicenter, randomized, double-blind, placebo-controlled trial conducted across 34 sites (fictional). Adults aged 18-75 with HbA1c between

7.5% and 10.5% on stable metformin dose ( $\geq 1500$  mg/day) for at least 8 weeks were randomized 1:1 to receive metformin/glipizide 1000 mg/5 mg twice daily (n=412)

or placebo (n=408), added to background metformin. Primary endpoint was change in

HbA1c from baseline to week 26. Secondary endpoints included proportion of patients achieving HbA1c < 7%, change in fasting plasma glucose, and change in body weight.

### RESULTS

Baseline mean HbA1c was 8.4% (SD 0.7) in both arms. At week 26, the metformin/glipizide group demonstrated a mean HbA1c reduction of 1.1% (95% CI

-1.2 to -0.9) versus 0.2% (95% CI -0.3 to -0.1) with placebo (between-group difference -0.9%,  $p < 0.001$ ). A significantly greater proportion of patients in the

treatment group achieved HbA1c < 7% (48.3% vs. 16.7%,  $p < 0.001$ ). Mean body weight change was -1.4 kg in the treatment group versus -0.3 kg with placebo. The most common adverse events in the treatment group were nausea (11.2%), diarrhea (9.8%), and headache (6.5%); rates of confirmed hypoglycemia were low (2.1% vs. 1.8% placebo). Three cases of pancreatitis were reported in the treatment group versus zero in placebo, prompting a recommendation for continued post-marketing surveillance.

### CONCLUSIONS

Metformin/glipizide combination therapy added to metformin significantly improved glycemic control compared to placebo over 26 weeks, with a manageable gastrointestinal adverse event profile. The observed pancreatitis signal, while based on a small number of cases, warrants further evaluation in larger, longer-duration studies.

### FUNDING (fictitious)

This study was funded by a fictional research grant and is provided solely as synthetic demonstration content; it does not represent a real clinical trial or real drug data.